

Module -1

DC Circuit Analysis and Theorems

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Reciprocity Theorem

In any passive linear bilateral network, if the single voltage source V_x in branch x produces the current response I_y in branch y , then the removal of the voltage source from branch x and its insertion in branch y will produce the current response I_y in branch x .

Another way of stating the theorem is to say that the interchange of an ideal voltage source and an ideal ammeter in any passive, linear, bilateral circuit will not change the ammeter reading.

Reciprocity Theorem

In any passive linear bilateral network, if the single current source I_x between nodes x and x' produces the voltage response V_y between nodes y and y' , then the removal of the current source from nodes x and x' and its insertion between nodes y and y' will produce the voltage response V_y between nodes x and x' .

In other words, the interchange of an ideal current source and an ideal voltmeter in any passive linear bilateral circuit will not change the voltmeter reading.

Reciprocity Theorem

According to the Reciprocity theorem statement the value of current obtained in any branch of an electrical circuit or network due to a single voltage source (V) in the circuit or network is the same as the value of current flowing through that branch when the source was originally connected and when the source was again connected to the branch where the value of current was originally determined.

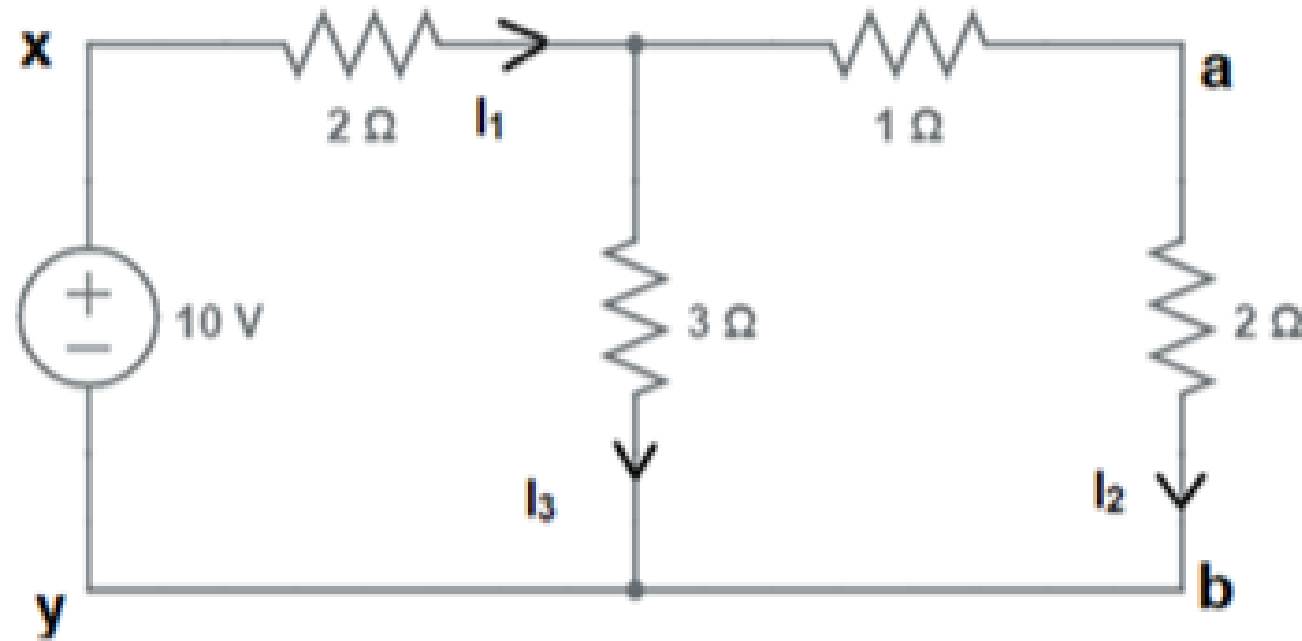
- The reciprocity theorem statement is used in both linear and bilateral networks where the circuit has just one independent source.
- It is utilized in both DC and AC circuits.

• **Steps to solve a Reciprocity Theorem**

- First, choose the branches in the circuit where reciprocity must be generated.
- Any normal network analysis approach may be used to determine the current flow within the branch.
- The voltage sources can be switched between the branches that have been chosen.
- Measuring current flow inside the branch, wherever the voltage source previously existed
- It is then noticed that the current obtained during the previous connection (i.e., the second step) and the flow of current measured after the source is exchanged (i.e., the fourth step) are equal.

Tutorial - 1

Test the reciprocity theorem at x-y and a-b.



Solution:

First calculate the current via the branch a-b as follows:

Equivalent resistance between x-y = $(3 \times 3) / 6 + 2 = 3.5 \Omega$

current $I_1 = 10 / 3.5 \text{ A}$

$= 2.86 \text{ A}$.

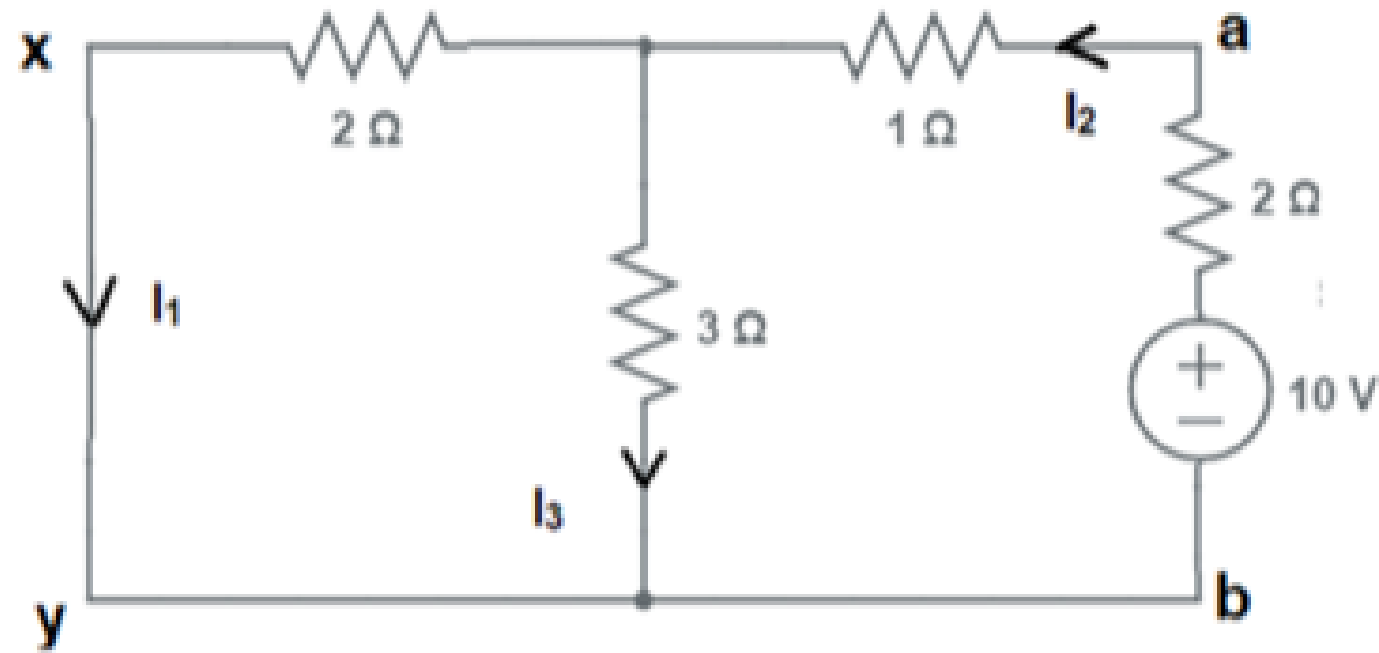
The current via branches a-b may now be estimated using the current division formula as follows:

$I_2 = (3 \times I_1) / (3 + 3)$

$= 2.86 / 2$

$= 1.43 \text{ A}$

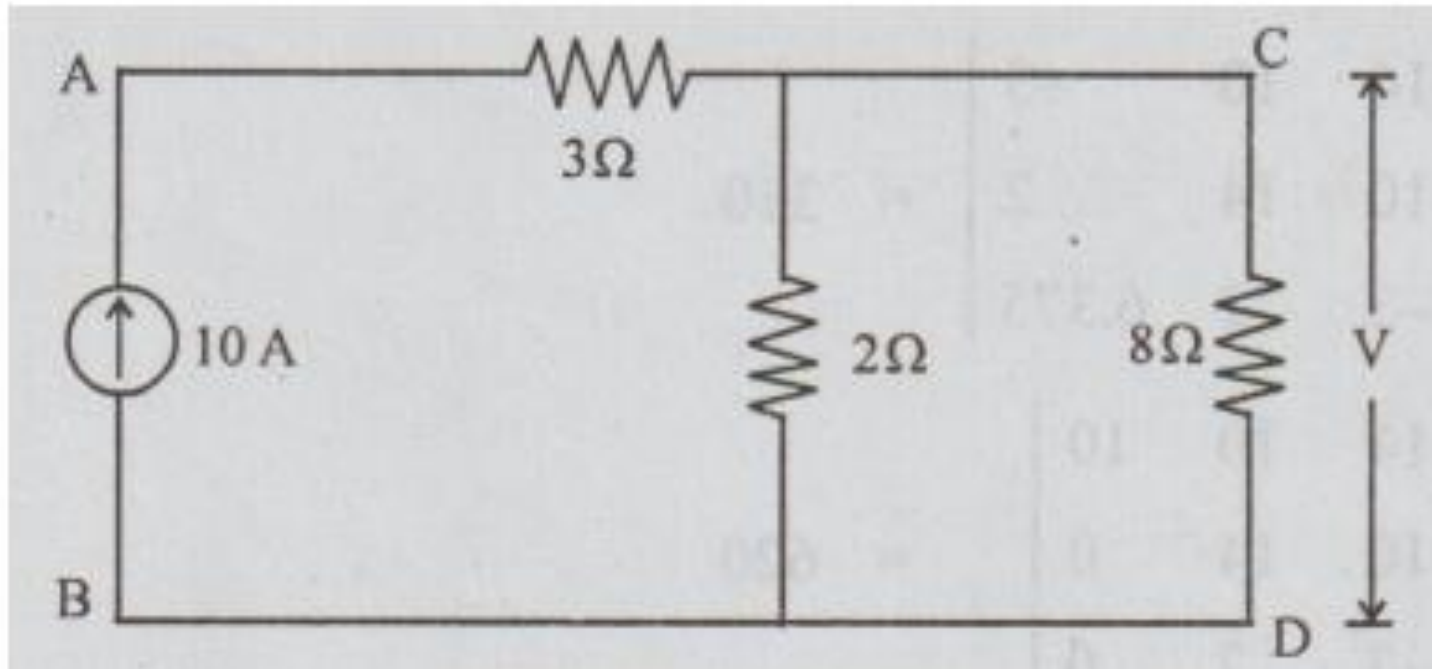
- Change the location of the voltage source and place it in branch a-b, as illustrated below:



- We want to determine the current value in branch x-y, which is the branch where the source was initially stored before being transferred to branch a-b, where we calculated the current.
- The following calculations are used to determine the equivalent resistance between terminals a-b:
 - $= 2 + 1 + 6/5$
 - $= 4.2\Omega$
 - Current $I_2 = 10/4.2$
 - $= 2.38\text{A}$
- Using the current division rule, the current in branch x-y may be calculated as follows:
 - $I_1 = (2.38 \times 3)/5$
 - $= 1.43\text{A}$
- As a result, I_1 and I_2 have the same value.

Tutorial - 2

Verify the reciprocity theorem in the circuit and calculate the transfer resistance.



Solution:

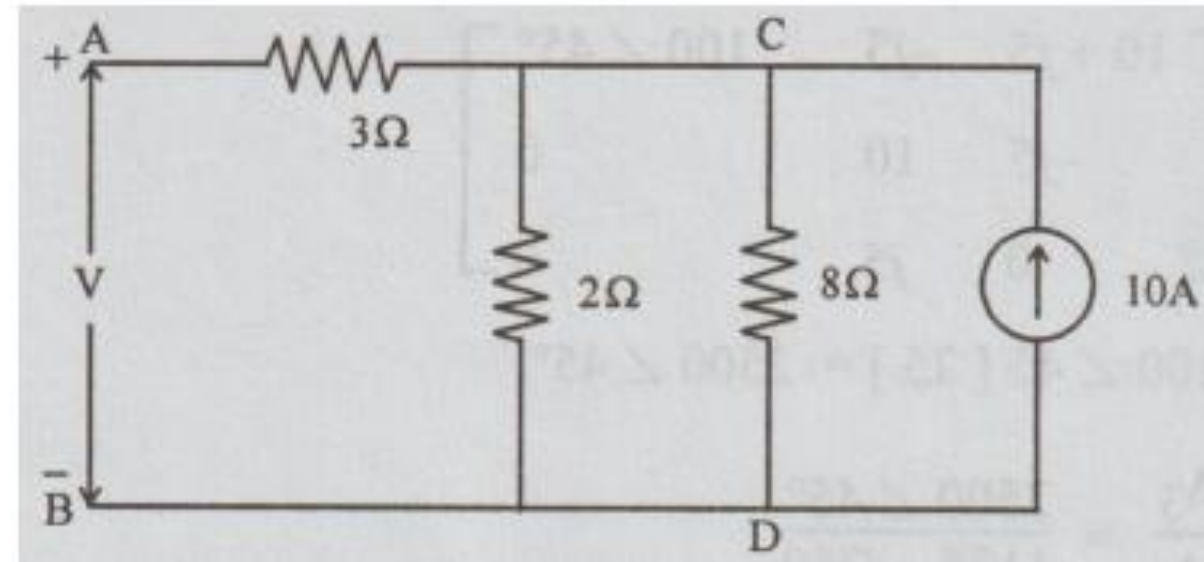
Step 1:

Voltage across 8 ohms = $V = I_g \times R$

But $I_g = 10 \times 2 / 2 + 8 = 2A$

$R = 8\Omega$

$V = 2 \times 8 = 16V$



Step 2: Interchanging the current source and response.

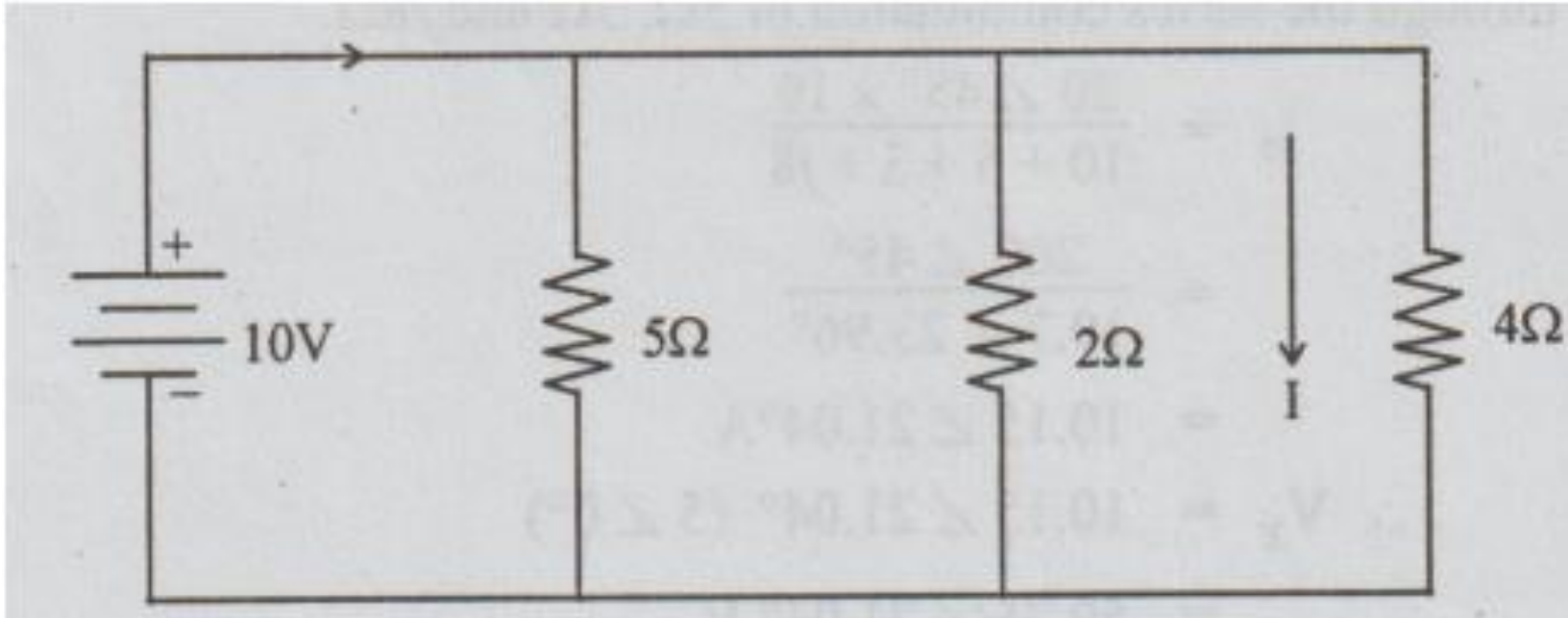
$V = I_2 \times R, R = 2\Omega,$

$V = 2 \times 8 = 16V$

The transfer resistance = $V / I = 16 / 10 = 1.6\Omega$

Tutorial - 3

Verify the reciprocity theorem in the circuit and calculate the transfer resistance.



Solution:

$$I = I_4$$
$$= 10 / 4 = 2.5 \text{ A}$$